Summary

1. Asheville-Buncombe Technical Community College (AB Tech)

Program Code: (Information Technology: Artificial Intelligence A. A. S.) A25590AI

Total Credit Hours: 64-65 credit hours

AB Tech structures its curriculum to combine core computer foundations (networking, programming, databases) with specialized AI coursework and general education.

Core Technical & AI Focus Areas

- **AI Foundations & Concepts:**
    - CSC 113– Artificial Intelligence Fundamentals
    - CSC 114– Artificial Intelligence I
    - CSC 117 -Artificial Intelligence Tools
    - ~~CSC 213– Artificial Intelligence & Decision Support Systems~~
- **Programming & Data Management:**
    - CSC 121– Python Programming
    - CSC 1211-Python Programming
    - DBA 120– Database Programming I (SQL)
    - DBA 220– Advanced Database Systems / Administration
- **IT Infrastructure & Security:**
    - CTI 110– Web, Programming, & DB Foundations
    - CTI 120– Network & Security Foundations
    - CTS 115– Information Systems Business Concepts
    - CTS 289– System Support Project (Capstone Project)
- **Ethics & Professional Skills:**
    - CTS 120– Hardware/Software Support

- CTS 240– Computer Ethics
- CTS 285– Systems Analysis & Design

2. Durham Technical Community College

Program Code: (Artificial Intelligence AAS) A25590R

Total Credit Hours: ~65-69 credit hours

1. Chatbot & Automation Programming

- **CSC 128– Chatbot Programming I**
  - **Scope:** Conversational interfaces, natural language processing (NLP) principles, and deploying intelligent software agents using industry-standard tools.
- **CSC 114– Artificial Intelligence Tools** *(or* ***– Artificial Intelligence I*** *depending on catalog year)* ***CSC 114***
  - **Scope:** Implementation of specialized software automation frameworks, API integration, and pre-built model tools.

2. Data Analytics

- **BAS 120– Introduction to Analytics**
  - **Scope:** Foundational methods for retrieving, structuring, visualizing, and interpreting data sets to support machine learning models and business decisions.
- **DBA 110– Database Concepts** *(or* ***– Database Programming I*** *)* ***DBA 120***
  - **Scope:** Relational database design, data manipulation, and structured query writing (SQL) required to feed data pipelines.
- **CSC 121 Python Programming**
  - **Scope: Python programming**

3. Ethical & Responsible AI Use

- **CSC 113– Artificial Intelligence Fundamentals**

    - **Scope:** Survey of artificial intelligence concepts, algorithms, real-world application models, and embedded modules on **AI ethics, bias, and responsible deployment** .

- **CSC 114– Artificial Intelligence 1**
    - **Scope: i**ntelligent agent design and rational decision making. Topics include goal-driven agents, search techniques, optimization.
- **CSC 161– Deep Learning**
    - **Scope:** Neural Networks
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- **CCT 231– Technology Crimes & Law** *(or core business ethics modules within* ***Information Systems Business Concepts*** *)* ***CTS 115***
    - **Scope:** Legal frameworks, digital security liability, copyright, governance, and societal impacts of emerging automation technologies.

3. Alamance Community College (ACC)

Program Code: (Artificial Intelligence AAS) A25710

(Note: ACC also integrates AI modules across its A25590IT pathways).

Total Credit Hours: 65-68 credit hours

ACC's curriculum places heavy emphasis on hands-on application, machine learning models, data analytics, and prompt engineering alongside standard IT support.

Core Technical & AI Focus Areas

- **AI & Machine Learning Core:**
    - CSC 113– Artificial Intelligence Fundamentals
    - CSC 114– Artificial intelligence tools
    - CSC 213– Applied Machine Learning & Intelligent Systems
    - CCT 231– Technology Crimes, AI & Law (Legal & Ethical Regulations)
- **Software Development & Data Science:**
    - CSC 121– Python Programming (Primary language used for AI model building)

- CSC 151– JAVA Programming or Object-Oriented Development
- DBA 120– Database Programming I
- NOS 110– Operating System Concepts

- **Systems & Network Foundation:**
  - CTI 110– Introduction to Information Technology
  - CTI 120– Network & Security Foundations
  - CTS 289– IT/AI Capstone Project